

Comparing Methods to Study Social Acceptance of Smart Glasses

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ABSTRACT

Conducting user studies on social perception of wearable technology requires a framework with sufficient sensitivity to emulate perception in real-life. Using videos can efficiently reach a large number of participants, but what level of video fidelity is necessary to achieve results similar to live interaction? In a study of 48 participants, we examine four different viewing conditions: phone, laptop, big screen, and live demonstration, while investigating the social weight of wearing different glasses (Vuzix Blade, Toozy Devkit, North Focals, or prescription glasses). Contrary to our hypothesis, the live condition was not the most sensitive.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Ubiquitous and mobile devices; Ubiquitous and mobile computing design and evaluation methods; Mixed / augmented reality.**

KEYWORDS

Head-worn displays; Smart Glasses; Social Acceptability; Social Perception; User Studies; Research Framework

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1 METHODS

After watching a demonstration performed by two actors (A and B), each participant (N=48) answers a questionnaire with eight Likert questions (e.g., “I would like to wear these glasses”) that are combined to form a social weight score. The demonstration consists of a 15-second coffee pantry conversation performed four times, each only differing by the type of glasses worn by actor A (Vuzix Blade, Toozy Devkit, North Focals, or prescription glasses). To establish a ground truth, 12 participants watched the demo live, and to study video fidelity, 12 participants each watched recordings of the same demo on a 6.06" mobile phone, a 13" laptop, and a 52" big screen where the actors appeared life-size. We plotted graphs for each viewing condition using these social weight scores. The y-axis represented social weight scores, and the x-axis represented the four glasses from least to most social weight, based on a pilot study we conducted that analyzed factors such as conspicuity of the glasses. As expected, we observed a general upward trend of acceptability in the graphs for each viewing condition, indicated by positive slopes.

2 RESULTS AND CONCLUSION

We expected that the live viewing condition would be the most discriminating (e.g., largest difference in scores between eyewear) resulting in the highest slope. For successively smaller and lower resolution displays, we expected the slope to be less steep. However, the slopes were inverse to that expected: live (4.04), big screen (4.11), laptop (4.13), and phone (4.34). No difference reached the level of statistical significance. All recorded conditions tended to be more discriminative than the live condition. We speculate that staring at the actors' technology during the live condition has a level of social opprobrium. However, when viewing a recording, participants may feel more comfortable constantly observing the actors and be more sensitive to visual cues of the wearable technology. In any case, the trend from the study suggests that if a device is deemed to be socially acceptable in a recording, then it is also likely acceptable during live observation. In other words, the results suggest that using video recordings for social weight experiments seems adequate for the task (at least for the college-age participants in this study).